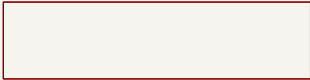


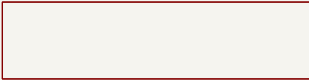
UNDER-DASH POWER BOARD: block diagram. Each box is a sheet (double-click it in KiCad).
Power rails +12V, +5V, +3V3 and GND join the sheets by name (power symbols);
every other connection between sheets is a wire here.

12 V input and protection



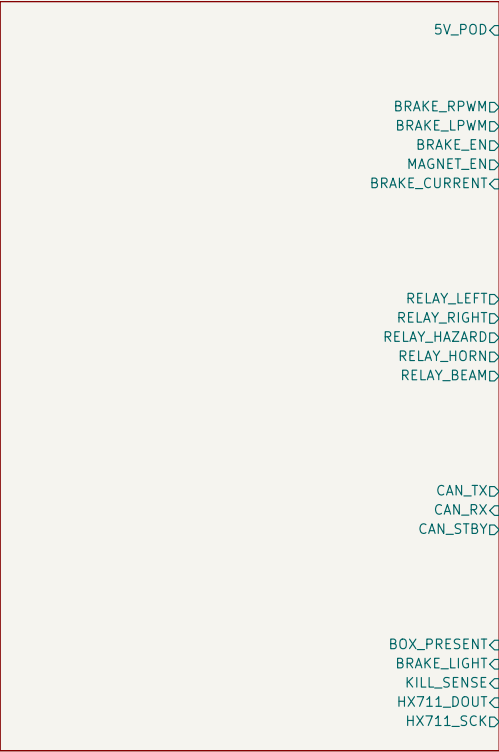
File: input.kicad_sch

5 V converter (TPS54360, TI reference design)



File: buck.kicad_sch

Inter-box cable: DB-25 to the pod

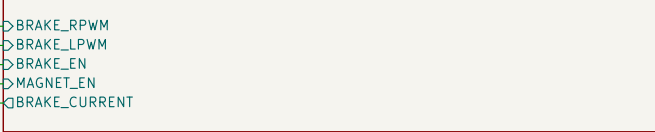


File: cable.kicad_sch

3.3 V, the pod's 5 V, the LIDAR feed

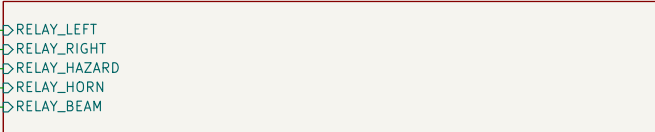


Brake, actuator, buffer, BTS7960, current sense, cable magnet



File: actuator.kicad_sch

Signal relays: indicators, hazards, horn, high beam



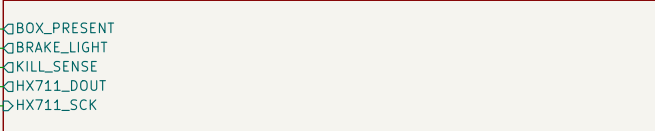
File: relays.kicad_sch

CAN transceiver for OBD-II



File: can.kicad_sch

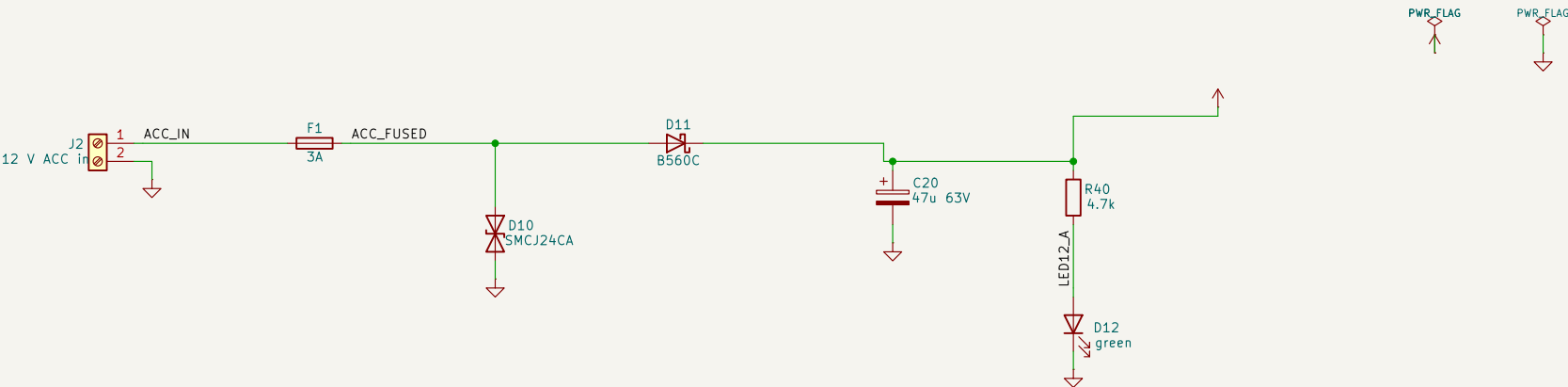
Presence, brake light, E-stop sense, kill relay, bench



File: status.kicad_sch

12 V input -> 5 V converter -> 3.3 V and the pod's 5 V.
The pod drives the actuator, relays and CAN;
the status sheet reports back to it.

12 V INPUT (the logic feed): switched ACC from the car's fuse box, so the board is off with the ignition. F1 3 A fuse. D10 bidirectional TVS clamps load-dump and jump-start spikes to about 39 V. D11 Schottky (60 V, 5 A) blocks a reversed lead. D12: +12V present.



Mounting holes: M3, no electrical connection

H1
M3

H2
M3

H3
M3

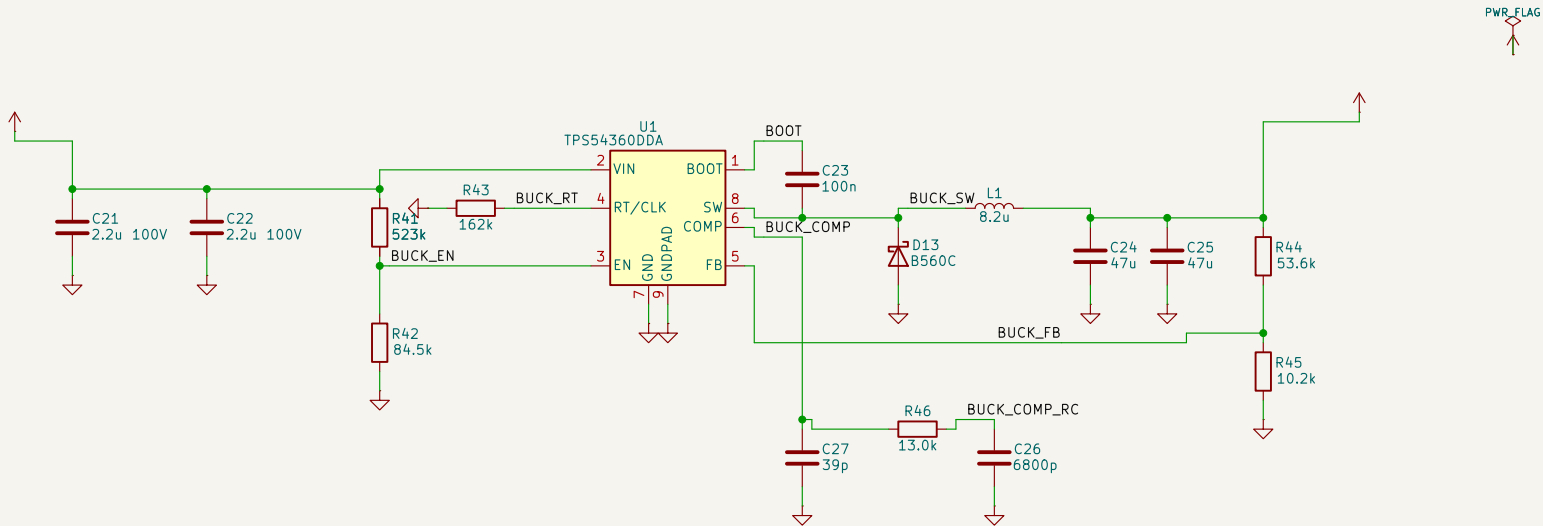
H4
M3

Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP – Kiogora Ian Mwenda
Sheet: /12 V input and protection/
File: input.kicad_sch

Title: 12 V input and protection

Size: A4	Date: 2026-10-07	Rev: 1
KiCad E.D.A. 10.0.1		Id: 2/9

5 V: TI's TPS54360 5 V / 3.5 A reference design (datasheet SLVSBB4G, Figure 34), unchanged.
Rated to 60 V input, so whatever the TVS lets through a load dump cannot destroy it.
R41/R42 start it at 8 V and stop it at 6.25 V. R43 sets 600 kHz. R44/R45 set 5.0 V.
C23 is the bootstrap for the high-side switch; D13 carries the current while it is off.
R46, C26, C27 compensate the control loop. Lay out C21/C22, U1, D13 and L1 as one tight loop.

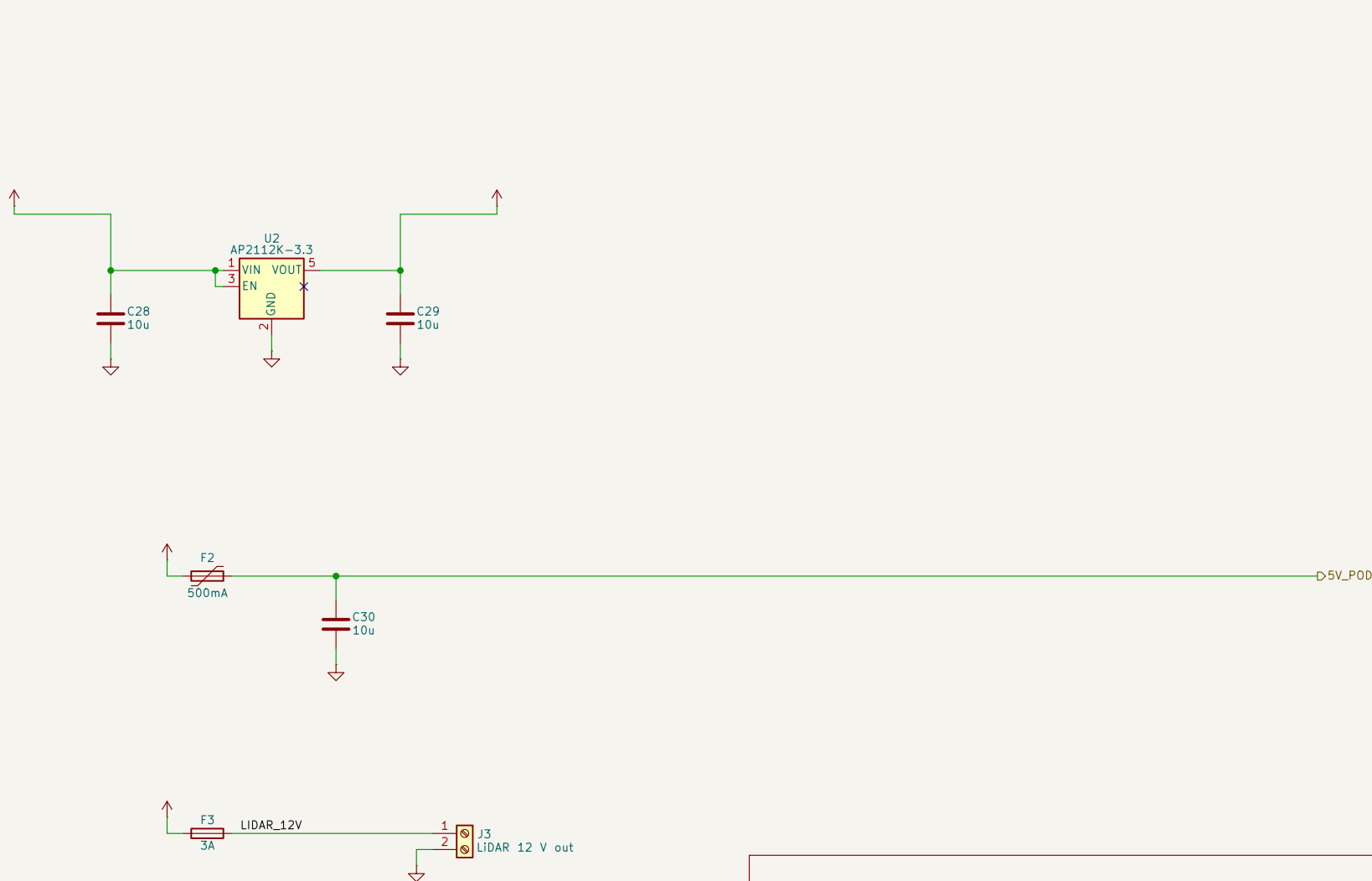


Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP – Kiogora Ian Mwenda
Sheet: /5 V converter (TPS54360, TI reference design)/
File: buck.kicad_sch

Title: 5 V converter (TPS54360, TI reference design)

Size: A4	Date: 2026-10-07	Rev: 1
KiCad E.D.A. 10.0.1		Id: 3/9

3.3 V (AP2112K) for the CAN transceiver only; the pod makes its own 3.3 V.
5V_POD: the pod's supply up the cable, through F2, a 500 mA resettable fuse,
so a short in the cable cannot take down this board's 5 V.
LiDAR: 12 V through its own 3 A fuse F3 to the M12 socket.



Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP – Kiogora Ian Mwenda
Sheet: /3.3 V, the pod's 5 V, the LiDAR feed/
File: rails.kicad_sch

Title: 3.3 V, the pod's 5 V, the LiDAR feed

Size: A4	Date: 2026-10-07	Rev: 1
KiCad E.D.A. 10.0.1		Id: 4/9

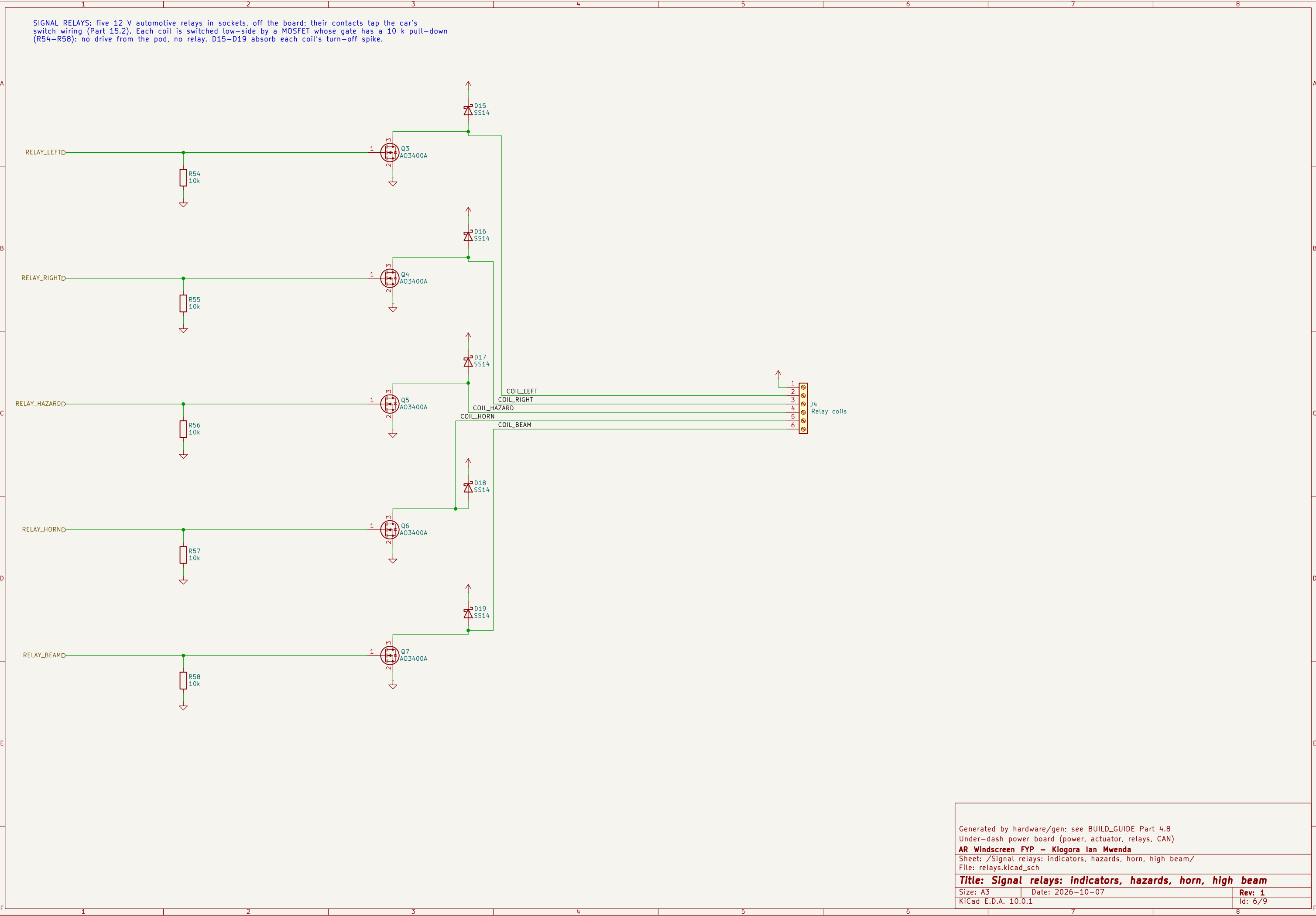
BRAKE ACTUATOR. Every line from the pod has a 10 k pull-down here (R50–R53): with the pod in reset, unpowered or unplugged, the drive, enable and magnet lines read LOW: released, cable dropped free. U3 (74HCT244, TTL thresholds) turns the pod's 3.3 V logic into the 5 V the BTS7960 module expects.

Actuator power comes ONLY through the 40 A kill relay's contacts (J5), whose coil is fed through the E-stop. Q2 holds the cable magnet; D14 absorbs its turn-off spike.

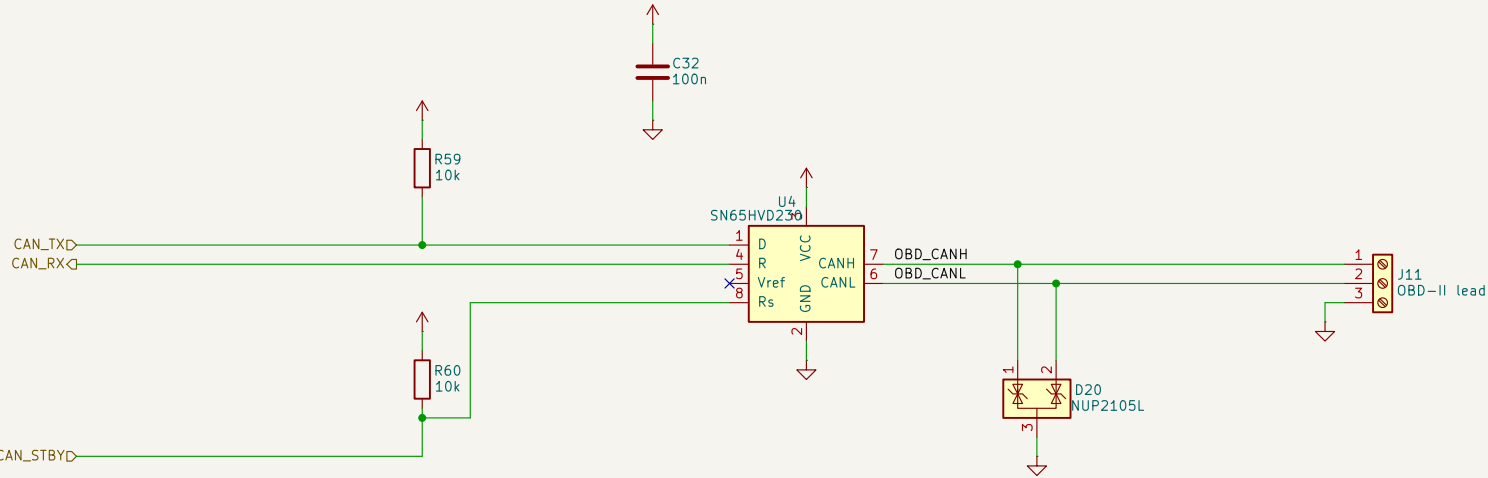
ACS712–20A module: 2.5 V at 0 A, 100 mV/A. R47/R48 scale it by 2/3 so nothing above 3.3 V travels up the cable to the pod's ADC.

Generated by hardware/gen; see BUILD_GUIDE Part 4.8 Under-dash power board (power, actuator, relays, CAN)		
AR Windscreen FYP – Klogora Ian Mwenda		
Sheet: /Brake actuator: buffer, BTS7960, current sense, cable magnet/ File: actuator.kicad_sch		
Title: Brake actuator: buffer, BTS7960, current sense, cable magnet		
Size: A3	Date: 2026-10-07	Rev: 1
KiCad E.D.A. 10.0.1		Id: 5/9

ACS712-20A module: 2.5 V at 0 A, 100 mV/A.
R47/R48 scale it by 2/3 so nothing above
3.3 V travels up the cable to the pod's ADC.



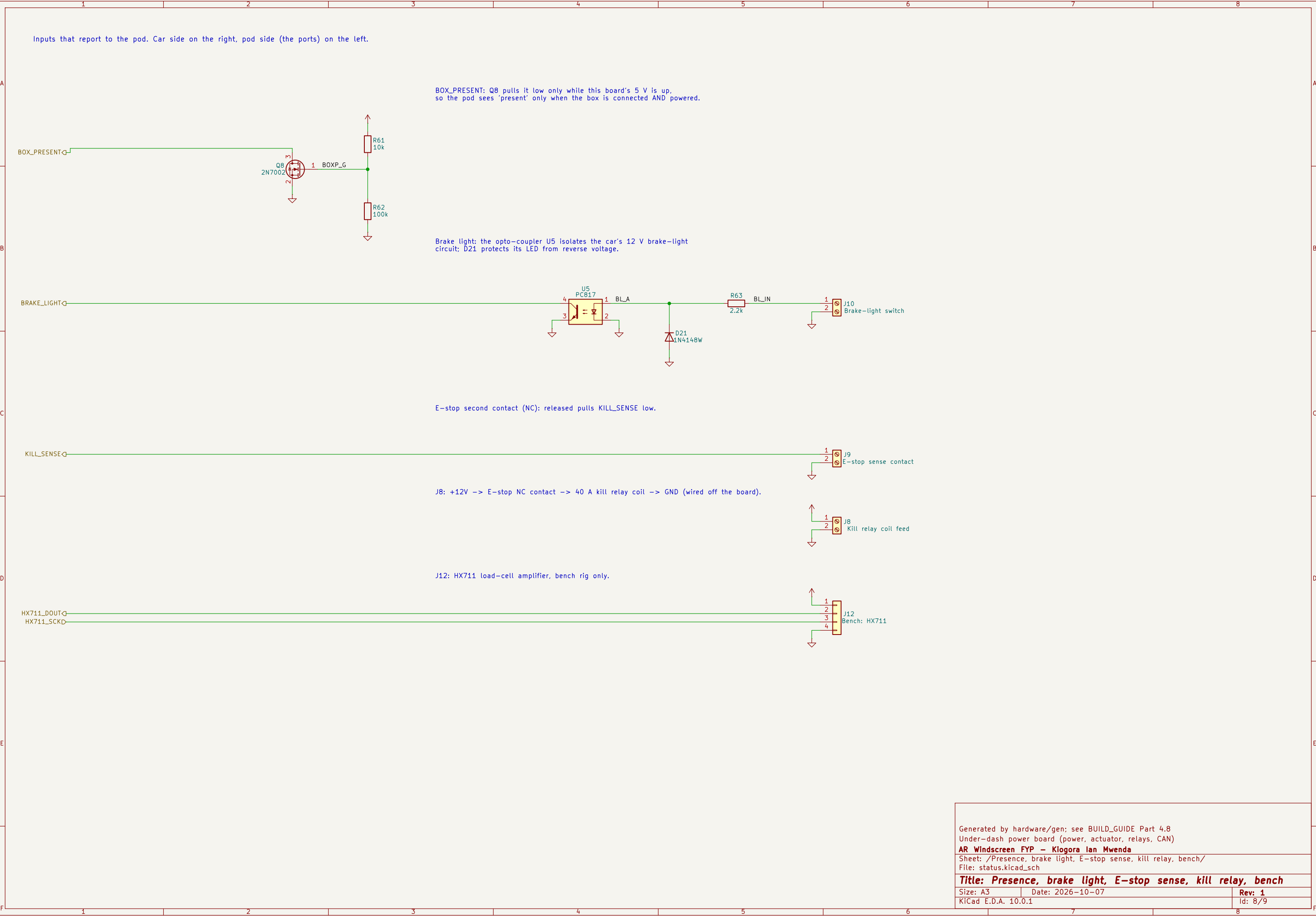
CAN (ISO 15765-4) to the car's OBD-II port, on this board so the unterminated branch off the car's bus is only the ~0.5 m OBD lead. R59 holds TX recessive and R60 holds Rs high (standby, cannot transmit) whenever the pod is not driving them. No termination: the car has its two.



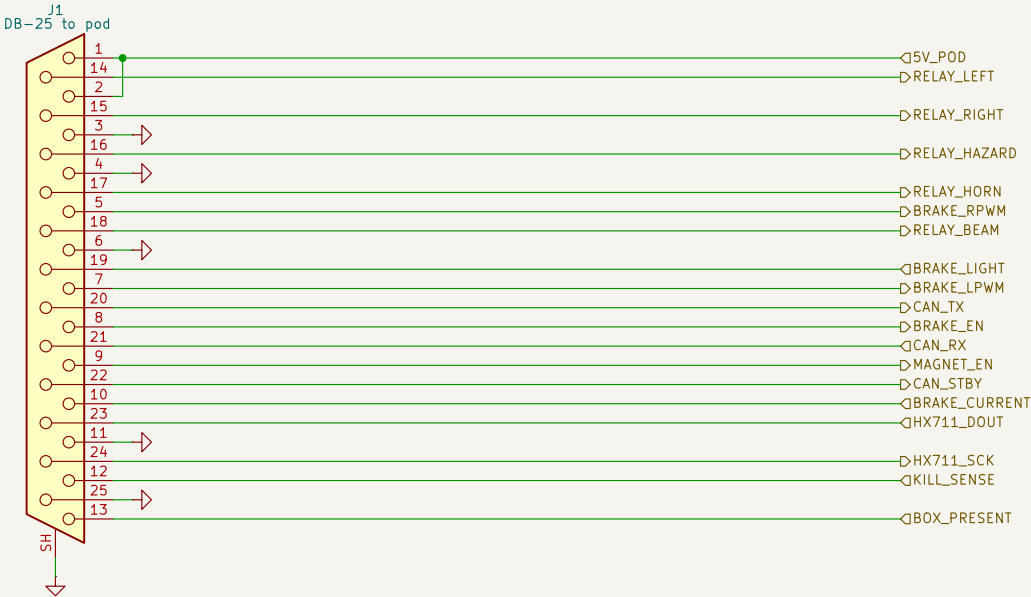
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP – Kiogora Ian Mwenda
Sheet: /CAN transceiver for OBD-II/
File: can.kicad_sch

Title: CAN transceiver for OBD-II

Size: A4	Date: 2026-10-07	Rev: 1
KiCad E.D.A. 10.0.1		Id: 7/9



INTER-BOX CABLE (BUILD_GUIDE 4.8.3). Shield bonded to GND at this end only.
Every line driven by the pod has its fail-safe pull-down on the sheet that uses it.



Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)

AR Windscreen FYP – Kiogora Ian Mwenda

Sheet: /Inter-box cable: DB-25 to the pod/
File: cable.kicad_sch

Title: Inter-box cable: DB-25 to the pod

Size: A4

Date: 2026-10-07

Rev: 1

KiCad E.D.A. 10.0.1

Id: 9/9